DONGKYU CHO

Computer Science Ph.D. Candidate at NYU — Research Intern at Microsoft Research

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EDUCATION

New York University — *New York, New York*
Ph.D. in Computer Science, advised by Prof. Rumi Chunara — *Sep. 2024 — May. 2028*
- Foundation Models, Continual Learning, Post-training (RL & Distillation)

Seoul National University — *Seoul, South Korea*
Master of Science in Data Science, advised by Prof. Sanghack Lee
- Causal Representation Learning, Domain Generalization

Seoul National University — *Seoul, South Korea*
Bachelor of Arts in History | *Information Science*
- Computational Linguistics & Historical Language Models. Graduated with Distinction (Cum Laude)

RESEARCH EXPERIENCE

Microsoft Corporation (Microsoft Research) — *Redmond, United States*
Research Intern — *May. 2026 — Aug. 2026*
- Research Area: Continual Learning, Self-Improvement, World Modeling
- Advisor: Dr. Ishai Menache, Dr. Sirui Li

Amazon.com Services LLC. (Amazon Science) — *Seattle, United States*
Applied Scientist Intern — *May. 2025 – Aug. 2025*
- Research Area: Reasoning and RLVR
- Advisor: Prof. Hengrui Cai, Dr. Rui Song, and Prof. Sheng Wang

LG AI Research — *Seoul, South Korea*
Research Scientist Intern — *Jul. 2023 – Jun. 2024*
- Research Area: Multi-modal Foundation Models (Time-Series/Language)
- Projects: Foundational Framework for Long-Term Time-Series Forecasting, Causal Discovery

Seoul National University — *Seoul, South Korea*
Graduate Research Assistant, advised by Prof. Sanghack Lee — *Jul. 2021 – Jul. 2023*
- Research Area: Causality, Domain Generalization
- Projects: Financial Language Models, Causal Language Models (with LG AI Research)

SELECTED PUBLICATIONS

*: co-first author

(C: Conference, W: Workshop, J: Journal, P: Preprint)

[C4] "Expert-guided Clinical Text Augmentation via Query-Based Model Collaboration "
Dongkyu Cho*, Miao Zhang*, Gregory D Lyng, Rumi Chunara.
Forty-Third International Conference on Machine Learning (ICML) 2026 — Poster

[C3] "Forget Forgetting: Continual Learning in a World of Abundant Memory"
Dongkyu Cho, Taesup Moon, Rumi Chunara, Kyunghyun Cho, Sungmin Cha,
International Conference on Learning Representations (ICLR) 2026 — Poster

[C2] "Tree of Concepts: Interpretable Continual Learners in Non-Stationary Clinical Domains"
Dongkyu Cho, Xiyue Li, Samrachana Adhikari, Rumi Chunara,
Machine Learning for Healthcare (MLHC) 2026 — Oral

[C1] "PEER Pressure: Model-to-Model Regularization for Single Source Domain Generalization"
Dongkyu Cho, Inwoo Hwang., Sanghack Lee,
IEEE/CVF Conference on Computer Vision & Pattern Recognition (CVPR) 2025 – Poster

[W3] "Correct Reasoning Paths Visit Shared Decision Pivots"
Dongkyu Cho, Amy B.Z. Zhang, Sheng Wang, Rumi, Chunara, Rui Song, Hengrui Cai.
NeurIPS 2025, Workshop on Foundations of Reasoning in Language Models (FoRLM)
NeurIPS 2025, Workshop on Efficient Reasoning (ER) – Spotlight

[W2] "ShERPA: Leveraging Neuron Alignment for Knowledge-preserving Fine-tuning",
Dongkyu Cho, Jinseok Yang, Jun Seo, Seohui Bae, Dongwan Kang, Soyeon Park, Hyeokjun Choe, Wonbin Ahn, Woohyung Lim.
ICLR 2024, Workshop on Mathematical and Empirical Understanding of Foundation Models (ME-FoMO)

[W1] "Learning to ignore: Single Source Domain Generalization via Oracle Regularization"
Dongkyu Cho, Sanghack Lee.,
NeurIPS 2023, Causal Representation Learning Workshop

[P4] "Self-Preservation in Language Models"
Dongkyu Cho, Rumi Chunara, Ali Behrouz.,
Preprint 2026

[P3] "Continual Learning via Self-Probe Gradients"
Dongkyu Cho, Rumi Chunara, Sungmin Cha.,
Preprint 2026

[P2] "Forgetting in Language Models: Capacity, Optimization, and Self-Generated Replay"
Martin Marek, **Dongkyu Cho**, Shikai Qiu, Rumi Chunara, Pavel Izmailov, Andrew Gordon Wilson,
Preprint 2026

[P1] "Do Clinical Models Change Treatment Decisions?"
Dongkyu Cho, Miao Zhang, Rumi Chunara.
Preprint 2026

TEACHING EXPERIENCE

Deep Learning and Machine Learning II, Teaching Assistant *Fall 2022*
- Assisted graduate course on statistical learning methods (e.g., Bayesian Frameworks).

Data Science Bootcamp, Teaching Assistant *Winter 2021*
- Assisted Undergraduate students with the introduction to machine learning.
- Lab sessions on machine learning libraries (e.g., Pytorch), Assignment design

INVITED TALKS

- Student Researcher Seminar, Graduate School of Data Science, Seoul National University (2023 June)
- Digital Health Group, New York University (2025 April)
- OpenAI X Center for Health Data Science, New York University (2025 September)

AWARD AND SCHOLARSHIP

- **School of Engineering (SoE) Fellowship**: New York University (2024)
- **NYU SGA Conference Grant**: New York University (2025, 2026)
- **ICLR Conference Grant**: ICLR (2026)

SERVICES AND ACTIVITIES

- **Reviewer**: NeurIPS (2026), ICML (2026), ICLR (2026), TPAMI (2026), CoLLAs (2025)
- **Languages**: English (Fluent), Korean (Native), French (Intermediate)
- **Test Scores**: TOEFL: 118 (2020.08), 116 (2022.08)